Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/16/2014
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	01/31/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	7
No. of Master's Candidates:	9	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 1/31/2014 per PI (Ed., 1/17/14) NOTE: Extended to 4/2/2014, per P. Downey/MSFC (Ed., 2/21/2012) NOTE: Received extension until 4/02/2012 per NSSC (Ed., 4/15/2011) NOTE: Grant may be extended, per P. Downey/MSFC (Ed., 2/18/2011)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Finalized parameter set development for Batch I FeCo flight alloy samples (#9 FeCrNi and #22 NdFeB) and proposed Batch II sample composition (#39 TiZrNi). Invited symposium sponsored by Jacqueline Etay, CNRS/SIMaP Laboratoire de Recherche sur les Matériaux, Université de Grenoble FR, June 5, 2013. Prepared flight plan for ESA Parabolic Flight September 2-13 in Merignac FR on FeCrNi flight alloy sample to test camera sensitivity and optimize operational settings for recalescence detection. Served as symposium organizer for American Society for Gravitational and Space Research (ASGSR) annual meeting held in parallel with ISPS-5 in Orlando FL 3-9 November 2013. Served as session chair. International Working Group 11-13 December 2013 at ESA-ESTEC in Noordwijk ND. IWG 9 involved evaluation of final dust limits for upcoming safety reviews, confirming sample processing order for Batch I, sample selection for Batch II, and proposing samples for Batch III. Graduated two MS students and six BS students.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Lee J, Matson DM, Binder S, Kolbe M, Herlach D, Hyers RW. "Magnetohydrodynamic modeling and experimental validation of convection inside electromagnetically levitated Co-Cu droplets." Metallurgical and Materials Transactions B. 2013 Dec. Online publication. http://dx.doi.org/10.1007/s11663-013-9995-5 , Dec-2013
Dissertations and Theses	Xiao X. (Xiao Xiao) "Magnetohydrodynamic Modeling of Convection inside Electromagnetically Levitated Molten Fe50Co50 Droplets." MS Thesis, Tufts University, May 2013. , May-2013